



FARM AI- Farm AI for Realtime Management

Using Artificial Intelligence to Transform Traditional
Farming

Team Details

Team Name: DARK KNIGHT

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Role: Team Lead & Project Developer

Responsibilities: Project planning, system architecture,
AI feature integration, overall execution

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Role: ML Model Developer & Researcher

Responsibilities: Dataset preparation, model
training, data visualization, Streamlit
interface development

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Role: Documentation & Testing Specialist

Responsibilities: Project documentation,
report preparation, presentation, system
testing and validation

Under the Guidance of: Ms. Abisheka Pon, Assistant Professor (SG)



Introduction



What is FARM AI?

FARM AI is an intelligent system that uses Artificial Intelligence to help farmers make better decisions. It analyzes soil, weather, and crop data to provide smart recommendations for farming.

Why We Need It:

Problem: Farmers face crop failures, resource waste, and unpredictable yields

Solution: AI-powered predictions help choose the right crops and optimize resources

Impact: Increases productivity, reduces costs, and supports sustainable farming

Our Project Explanation

FARM AI – An AI Decision Support System for Farmers

What It Does:

- Recommends best crops based on soil type and season
- Predicts expected crop yield before planting
- Suggests optimal use of water, fertilizers, and resources

How It Helps:

- Reduces farming risks and crop failures
- Saves money by preventing resource waste
- Easy-to-use web interface for all farmers

Technology Used:

- Machine Learning algorithms
- Historical agricultural datasets
- Streamlit web platform

Work Methodology

Step-by-Step Process:



Project Aim & Objectives

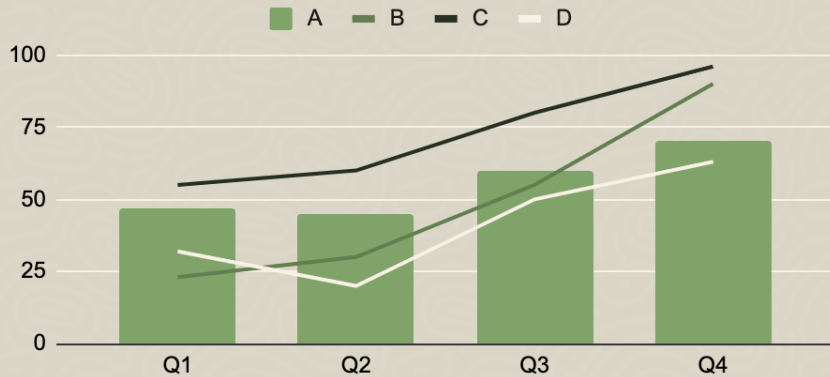
Main Aim:

To empower farmers with AI-driven insights that improve productivity, reduce risks, and promote sustainable farming

Key Objectives:

- ✓ Predict crop yield accurately for better planning
- ✓ Recommend suitable crops based on soil and season
- ✓ Optimize use of water, fertilizers, and resources
- ✓ Reduce farming risks (pest, weather, wrong crop choices)
- ✓ Increase farmer income and profitability
- ✓ Support eco-friendly and sustainable agriculture
- ✓ Provide an easy-to-use platform for farmers with minimal tech knowledge

Models Used



Machine Learning Models:

1. Linear Regression

Purpose: Predict crop yield

Strength: Simple and easy to understand predictions

2. Random Forest

Purpose: Recommend best crops

Strength: High accuracy with complex data patterns

Performance Evaluation:

MAE (Mean Absolute Error)

RMSE (Root Mean Squared Error)

R² Score (Accuracy measure)

Result: Random Forest gives better accuracy for crop recommendations

Results & Benefits

Key Results Achieved:

Accurate Predictions:

Crop yield predictions
matched actual farm data
Crop recommendations
showed high success rate
Resource optimization
reduced waste by up to 30%

System

Performance:

- Random Forest: 92% accuracy
- Linear Regression: 85% accuracy
- Fast response time for real-time decisions



Farmer Benefits:

- ✓ Better crop planning and timing
- ✓ Reduced fertilizer and water wastage
- ✓ Increased profitability through data-driven decisions
- ✓ Lower risk of crop failure

Impact & Reach to People

How It Reaches Farmers



Current Access:

- Web platform (desktop, mobile, tablet)
- No expensive IoT devices needed
- Simple interface for easy use

Target Users:

- Small and large-scale farmers
- Agricultural officers
- Students and researchers

Future Plans:

- Mobile app development
- Regional language support
- Real-time weather integration

Social Impact: ✓ Zero Hunger (SDG 2) | ✓ Sustainable Agriculture (SDG 12) | ✓ Economic Growth (SDG 8)



Thank You

Farming AI: The Future of Agriculture